Exponential and Logarithmic Equations

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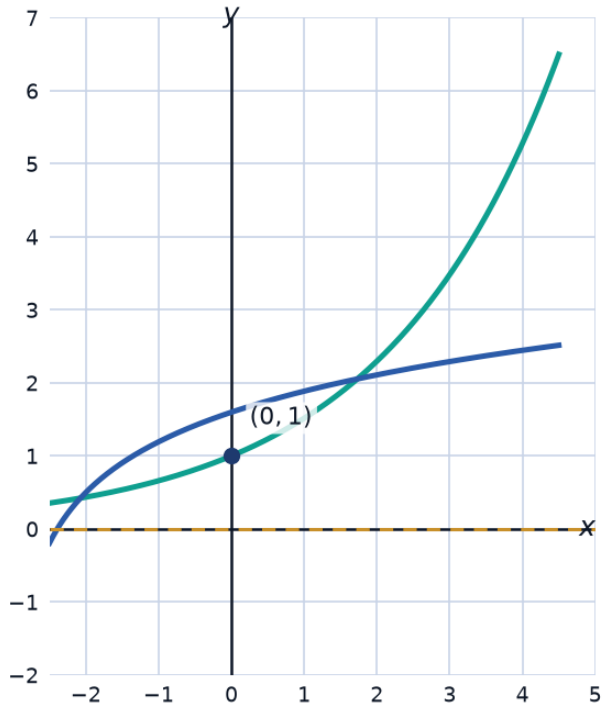
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Exponential equations

Solve $3^{x+1} = 81$ exactly.

2 [Foundational] • Target: Exponential equations

Solve $5^{2x} = 17$. Give an exact answer and a decimal to three places.



3 [Foundational] • Target: Exponential equations

Solve $e^{2x} - 5e^{x} + 6 = 0$ exactly.

4 [Foundational] • Target: Logarithmic equations

Solve $\ln(2x - 3) = 1$. Give an exact answer and a decimal to three places.

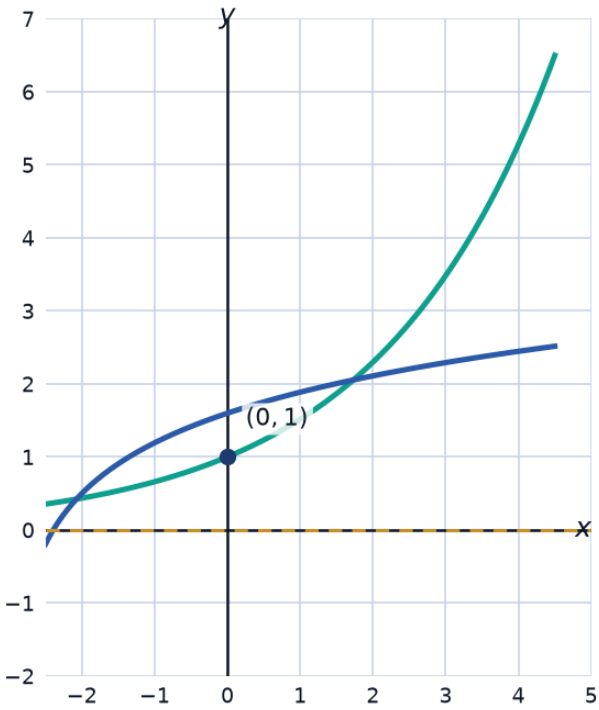
5 [Foundational] • Target: Logarithmic equations

Solve $\log_2 x + \log_2 (x - 6) = 4$, checking for extraneous solutions.

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6 [Foundational] • Target: Logarithmic equations

Solve $\log(x+3) - \log(x-1) = 1$, checking for extraneous solutions.



7 [Foundational] • Target: Logarithmic equations

The pH of a solution is $\text{pH} = -\log[\text{H}^+]$. Find the hydrogen-ion concentration $[\text{H}^+]$ of a solution with pH 4.6, in scientific notation to two significant figures.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: More exponential equations

Solve $2^x = 3^{x-1}$, giving an exact expression and a decimal to three places.

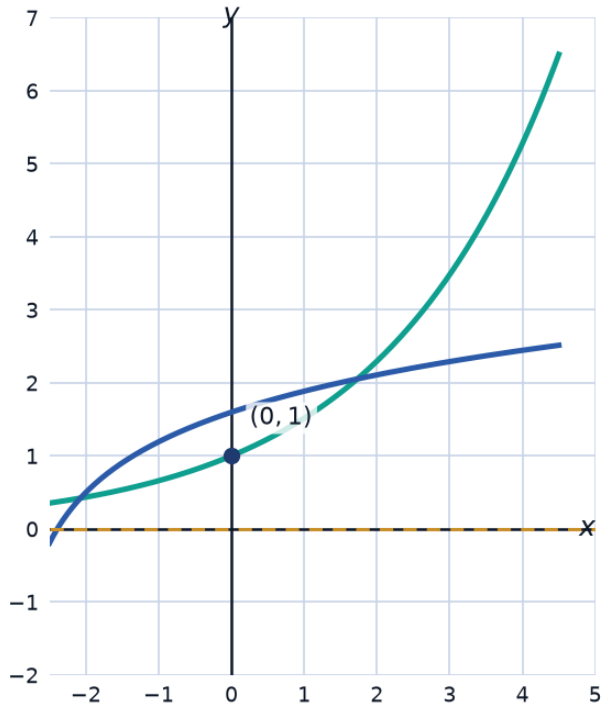
9 [Intermediate] • Target: More exponential equations

Solve $4^x - 3 \cdot 2^x - 4 = 0$ exactly, using the substitution $u = 2^x$.

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10 [Intermediate] • Target: Systems and applications

A bacteria culture grows according to $N(t) = 500e^{kt}$. If the population reaches 2000 after 6 hours, find k (4 d.p.) and hence the time for the population to reach 10,000, to one decimal place.



11 [Intermediate] • Target: Systems and applications

Solve the equation $e^x + e^{-x} = 3$ exactly, using the substitution $u = e^x$ (this expression is related to $2\cosh x$).

12 [Intermediate] • Target: Combined logarithmic and exponential practice

Solve $\log_3(x+2) + \log_3(x-2) = 2$, checking for extraneous solutions.

13 [Intermediate] • Target: Combined logarithmic and exponential practice

Solve $9^x = 27^{x-1}$ exactly, writing both sides with base 3.

14 [Intermediate] • Target: Equations requiring both exponential and logarithmic techniques

Solve $\log_2(x) = 3 - \log_2(x-2)$, checking for extraneous solutions.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

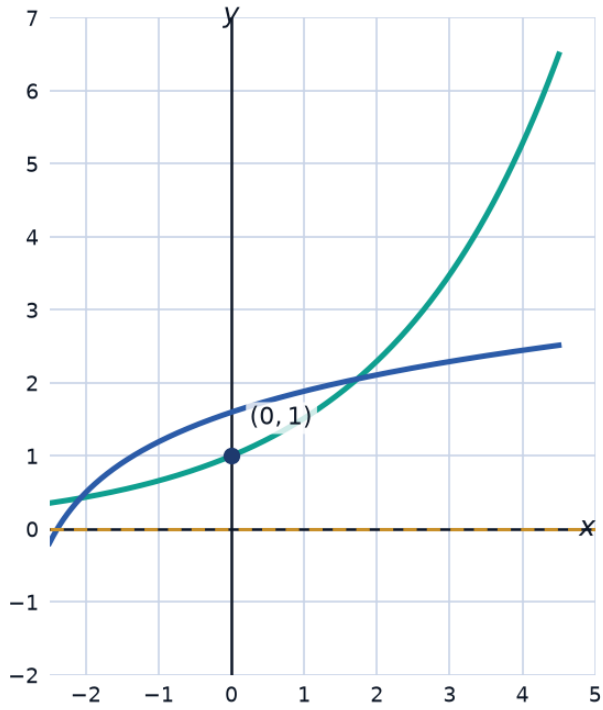
15 [Challenging] • Target: Equations requiring both exponential and logarithmic techniques • Cross-Lesson: 3.01 The Unit Circle and Exact Values

Solve $e^{\ln x + \ln 2} = 10$ for x, simplifying the exponent first.

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16 [Challenging] • Target: Verifying solutions numerically • Cross-Lesson: 3.02 Fundamental Trigonometric Identities

For the equation $5^{2x} = 17$ solved earlier ($x \approx 0.880$), verify the solution by substituting back and computing both sides to 2 decimal places.



17 [Challenging] • Target: Equations with different bases on each side • Cross-Lesson: 3.03 Sum and Difference Formulas

Solve $7^x = 3^{x+2}$, giving an exact expression and a decimal to three places.

18 [Challenging] • Target: Equations with different bases on each side • Cross-Lesson: 3.04 Double-Angle Formulas and Trigonometric Equations

Solve $10^{2x-1} = 500$, giving an exact logarithmic expression and a decimal to three places.

19 [Challenging] • Target: A modeling problem requiring both equation types • Cross-Lesson: 4.01 Circles and Parabolas

A population model is $P(t) = 800e^{0.05t}$. Find the time t when $P(t) = 2000$, to two decimal places, and separately find $P^{-1}(1500)$ (the time when the population reaches 1500), to two decimal places.

20 [Challenging] • Target: Checking solutions by substitution • Cross-Lesson: 4.02 Ellipses

Verify the solution $x = 8$ found earlier for $\log_2 x + \log_2 (x - 6) = 4$ by substituting back into the original equation.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Equations with Like Bases

Solve $27^{x-1} = 9^{2x+3}$ for x.

a. -9

b. 9

c. -3

d. 6

22 [Foundational] • Target: Equations with Like Bases

Solve $5^{x^2-3x} = 625$.

a. $x = -1$ and $x = 4$

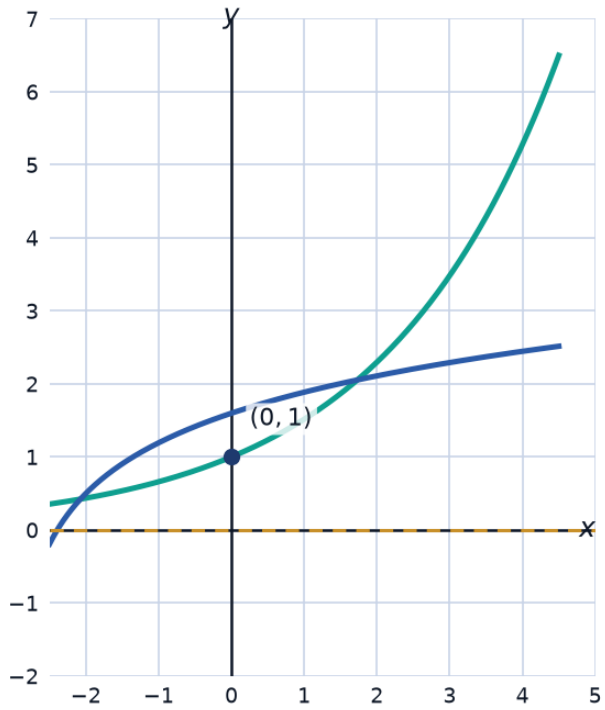
b. $x = 1$ and $x = -4$

c. $x = 2$ and $x = -2$

d. $x = 4$ only

23 [Foundational] • Target: Equations with Like Bases

Solve $\left(\frac{1}{2}\right)^{x+2} = 8^{x-1}$.



a. 1/4

b. 1/2

c. $-1/4$

d. 2

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24 [Foundational] • Target: Equations Requiring Natural Logarithms

Solve $3^{2x} = 7$ for x in exact form.

a. $x = \frac{\ln 7}{2\ln 3}$

b. $x = \frac{\ln 3}{2\ln 7}$

c. $x = \frac{7}{6}$

d. $x = \frac{\ln 7}{\ln 6}$

25 [Foundational] • Target: Equations Requiring Natural Logarithms

Solve $5e^{3x-2} = 40$ for x.

a. $x = \frac{\ln 8 + 2}{3}$

b. $x = \frac{\ln 8 - 2}{3}$

c. $x = \frac{\ln 40}{15}$

d. $x = 2$

26 [Foundational] • Target: Equations Requiring Natural Logarithms

Solve $2^{x+1} = 5^{x-1}$ for x.

a. $x = \frac{\ln 2 + \ln 5}{\ln 5 - \ln 2} = \frac{\ln 10}{\ln(5/2)}$

b. $x = \frac{\ln 5}{\ln 2}$

c. $x = 1$

d. $x = \frac{\ln 10}{\ln 10} = 1$

27 [Foundational] • Target: Logarithmic Equations and Extraneous Solutions

Solve $\log_2(x) + \log_2(x-2) = 3$.

a. $x = 4$ only

b. $x = 4$ and $x = -2$

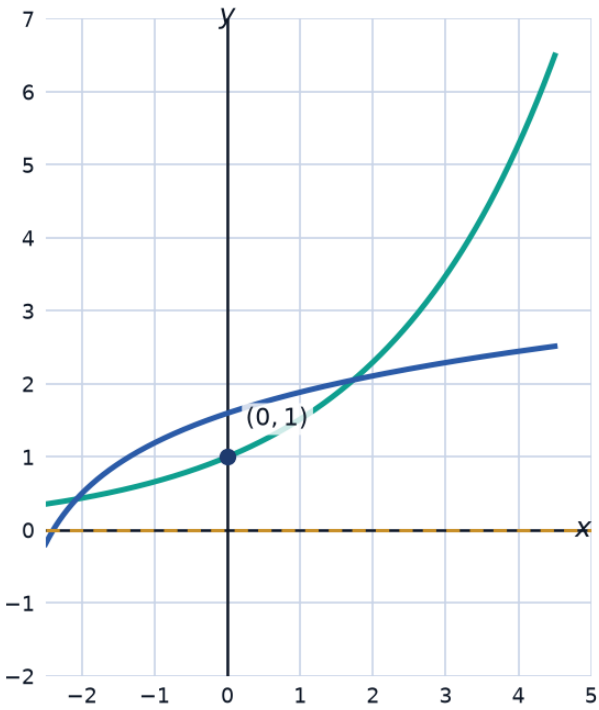
c. $x = 8$

d. $x = 2$

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28 [Foundational] • Target: Logarithmic Equations and Extraneous Solutions

Solve $\ln(x+5) - \ln(x-1) = \ln(3)$.



a. $x = 4$

b. $x = 3$

c. $x = 2$

d. $x = 5$

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Logarithmic Equations and Extraneous Solutions

Why must potential solutions to logarithmic equations always be checked in the original equation?

a. The argument of every logarithm must be strictly positive (> 0)

b. Logarithms cannot have rational answers

c. Extraneous roots always equal 0

d. Base change invalidates solutions

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30 [Intermediate] • Target: Quadratic-Form Exponential Equations

Solve $e^{2x} - 5e^{x} + 6 = 0$ for x.

a. $x = \ln 2$ and $x = \ln 3$

b. $x = 2$ and $x = 3$

c. $x = \ln 5$ and $x = \ln 6$

d. $x = e^{2}$ and $x = e^{3}$

31 [Intermediate] • Target: Quadratic-Form Exponential Equations

Solve $e^{2x} - 3e^{x} - 4 = 0$ for x.

a. $x = \ln 4$ only

b. $x = \ln 4$ and $x = \ln(-1)$

c. $x = 4$ and $x = -1$

d. No real solutions

32 [Intermediate] • Target: Quadratic-Form Exponential Equations

Solve $4^{x} - 3 \cdot 2^{x} - 4 = 0$.

a. $x = 2$ only

b. $x = 4$ only

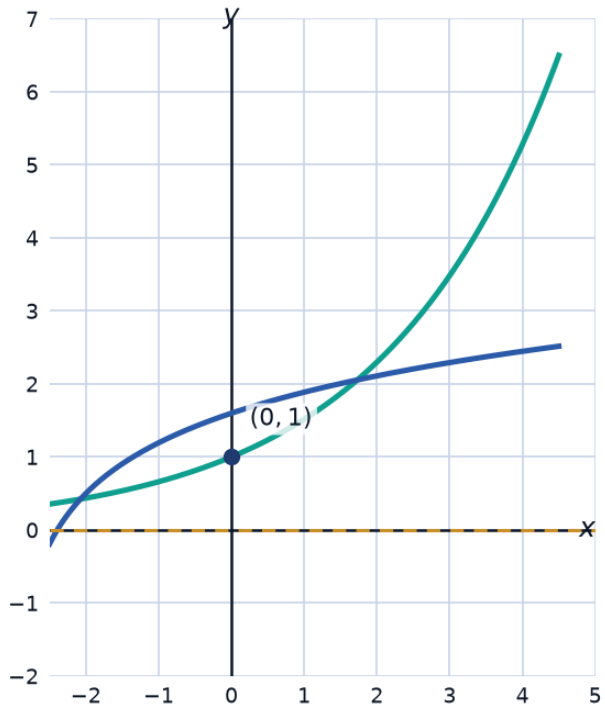
c. $x = 2$ and $x = -1$

d. $x = 1$

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33 [Intermediate] • Target: Logarithmic Equations and Extraneous Solutions

Solve $7^x = 100$ for x, expressed using logarithms.



a. $x = \frac{100}{7}$

b. $x = 100 - 7$

c. $x = \log_7(100) = \frac{\ln 100}{\ln 7}$

d. $x = \log_{100}(7)$

34 [Intermediate] • Target: Quadratic-Form Exponential Equations

Solve $7^x = 10$ for x, expressed using logarithms.

a. $x = 10 - 7$

b. $x = \log_7(10) = \frac{\ln 10}{\ln 7}$

c. $x = \frac{10}{7}$

d. $x = \log_{10}(7)$

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35 **[Intermediate]** • **Target: Exponential equations**

Solve $3^x = 50$ for x, expressed using logarithms.

a. $x = \log_{50}(3)$

b. $x = \frac{50}{3}$

c. $x = \log_3(50) = \frac{\ln 50}{\ln 3}$

d. $x = 50 - 3$

36 **[Intermediate]** • **Target: Logarithmic equations**

Solve $5^x = 20$ for x, expressed using logarithms.

a. $x = \log_5(20) = \frac{\ln 20}{\ln 5}$

b. $x = \frac{20}{5}$

c. $x = \log_{20}(5)$

d. $x = 20 - 5$

37 **[Intermediate]** • **Target: More exponential equations**

Solve $2^x = 15$ for x, expressed using logarithms.

a. $x = \frac{15}{2}$

b. $x = 15 - 2$

c. $x = \log_2(15) = \frac{\ln 15}{\ln 2}$

d. $x = \log_{15}(2)$

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

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38 [Challenging] • Target: Systems and applications • Cross-Lesson: 3.04 Double-Angle Formulas and Trigonometric Equations

Solve $3^x = 20$ for x, expressed using logarithms.

a. $x = \log_{20}(3)$

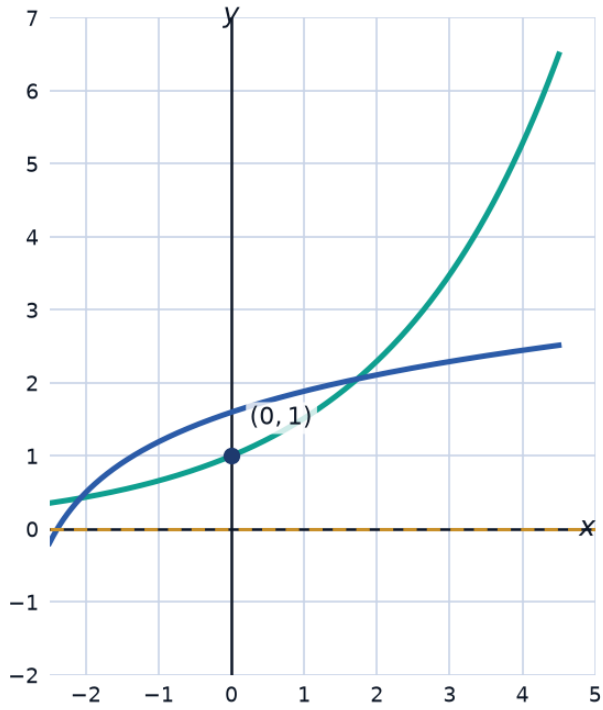
b. $x = \frac{20}{3}$

c. $x = 20 - 3$

d. $x = \log_3(20) = \frac{\ln 20}{\ln 3}$

39 [Challenging] • Target: Combined logarithmic and exponential practice • Cross-Lesson: 4.01 Circles and Parabolas

Solve $2^x = 20$ for x, expressed using logarithms.



a. $x = \log_{20}(2)$

b. $x = 20 - 2$

c. $x = \frac{20}{2}$

d. $x = \log_2(20) = \frac{\ln 20}{\ln 2}$

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40 [Challenging] • Target: Equations requiring both exponential and logarithmic techniques • Cross-Lesson: 4.02 Ellipses

Solve $7^x = 15$ for x, expressed using logarithms.

a. $x = \frac{15}{7}$

b. $x = 15 - 7$

c. $x = \log_{15}(7)$

d. $x = \log_7(15) = \frac{\ln 15}{\ln 7}$

41 [Challenging] • Target: Verifying solutions numerically • Cross-Lesson: 4.03 Hyperbolas

Solve $3^x = 10$ for x, expressed using logarithms.

a. $x = \log_3(10) = \frac{\ln 10}{\ln 3}$

b. $x = 10 - 3$

c. $x = \log_{10}(3)$

d. $x = \frac{10}{3}$

42 [Challenging] • Target: Equations with different bases on each side • Cross-Lesson: 5.01 Vectors: Components and Operations

Solve $3^x = 15$ for x, expressed using logarithms.

a. $x = \log_{15}(3)$

b. $x = \frac{15}{3}$

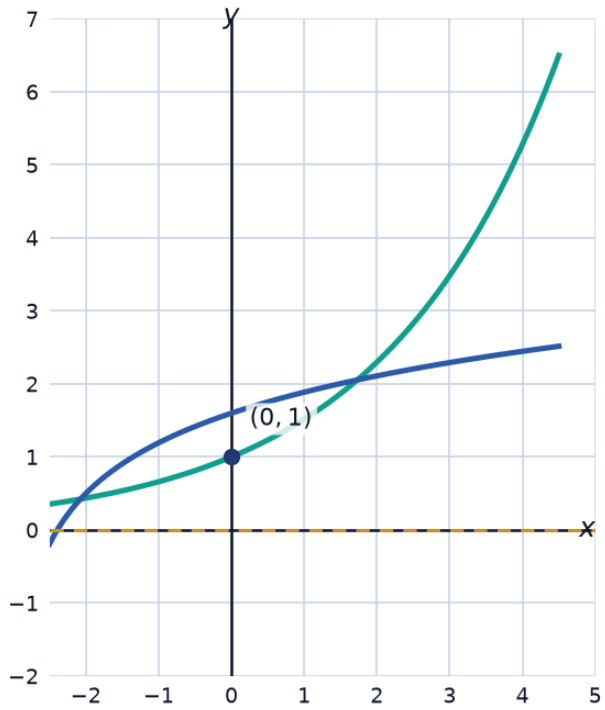
c. $x = \log_3(15) = \frac{\ln 15}{\ln 3}$

d. $x = 15 - 3$

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43 [Challenging] • Target: A modeling problem requiring both equation types • Cross-Lesson: 5.02 The Dot Product

Solve $7^x = 50$ for x, expressed using logarithms.



a. $x = \frac{50}{7}$

b. $x = \log_7(50) = \frac{\ln 50}{\ln 7}$

c. $x = \log_{50}(7)$

d. $x = 50 - 7$

44 [Challenging] • Target: Checking solutions by substitution • Cross-Lesson: 5.03 Parametric Equations

Write $7^4 = 2401$ in logarithmic form.

a. $\log_7(4) = 2401$

b. $\log_7(2401) = 4$

c. $\log_4(7) = 2401$

d. $\log_{2401}(7) = 4$

45 [Challenging] • Target: Equations with Like Bases • Cross-Lesson: 6.01 Polar Coordinates and Conversions

Write $5^4 = 625$ in logarithmic form.

a. $\log_{625}(5) = 4$

b. $\log_5(4) = 625$

c. $\log_4(5) = 625$

d. $\log_5(625) = 4$

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